

# SHACHINDRAN A/L VEERANGAN

Information Technology Undergraduate | AI/NLP · Backend Engineering · Applied Machine Learning

Kuantan, Malaysia | +60 19-958 9799 | shachindran4@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## EDUCATION

### Universiti Teknologi PETRONAS (UTP) — Bachelor of Information Technology (Hons)

May 2024 – Expected Aug 2027

Malaysia | Current CGPA: 3.41/4.00 | Dean's List: 3 of 6 completed semesters

### Bilkent University — Exchange Student, Computer Information Systems and Technologies (CTIS)

Sep 2025 – Jan 2026

Ankara, Türkiye

### Universiti Teknologi PETRONAS (UTP) — Foundation in Information Technology

2023 – 2024

Malaysia | CGPA: 3.78/4.00 | Dean's List in all three semesters | CFS Dean's Award

## RESEARCH & TECHNICAL EXPERIENCE

### Omniscien Technologies — Artificial Intelligence & Natural Language Processing Intern

May 2026 – Present

Bangkok, Thailand

- Expanded document routing and metadata support from 28 to 100 file formats, independently validating 84 end-to-end across office, archive, image, geospatial, medical and specialist formats; the work was merged into the project's development repository.
- Benchmarked IBM Docling against Unstructured on complex PDFs containing tables, formulas and multi-column layouts, then used the findings to build hybrid routing between faster and higher-fidelity extraction paths.
- Investigated a customer-reported Java/Aspose.PDF sanitization defect, traced the watermark to a vector Form XObject and implemented a fix that removed it from the original sample while preserving legitimate document content.
- Separated GPU-bound vision-language inference from CPU-bound extraction across isolated containerized workers, enabling routing between conventional OCR and local/remote multimodal analysis with structured-output validation and fallback handling for malformed responses.
- Expanded email ingestion and normalization to 11 validated formats and redesigned the pipeline around a canonical intermediate representation to remove duplicate extraction; the revised pipeline passed 97/97 automated tests.
- Diagnosed and resolved reliability issues across the document-processing pipeline, including a dependency conflict between OCR-related libraries and repeated model-initialization overhead in isolated worker processes.

## SELECTED TECHNICAL PROJECTS

### Predictive Maintenance for Turbofan Engines — Independent Machine Learning Project

2026

Python, scikit-learn, Pandas, NumPy | NASA C-MAPSS | [GitHub](#)

- Built an end-to-end Remaining Useful Life prediction workflow using NASA C-MAPSS turbofan sensor data, covering preprocessing, feature preparation, Random Forest regression and evaluation on held-out engines.
- Compared predicted and actual degradation trajectories and documented the workflow, results and visualizations for reproducibility.

### Smart Energy Consumption Forecaster — Independent Time-Series Project

2026

Python, Prophet, Pandas, NumPy | [GitHub](#)

- Built a forecasting workflow using hourly American Electric Power demand data from 2004–2018, modelling electricity consumption with Prophet.
- Analyzed daily, weekly and yearly seasonality and produced forecast/component visualizations to interpret recurring demand patterns and longer-term trends.

### **Wan and Whiskers — Game Development Project**

2026

Unity, C# | 5-person team

- Co-developed a two-player Unity action game, contributing approximately 50% of the C# codebase and implementing multi-character control, destructible environments and endless-mode systems.

### **Bank Loan Predictive Modeling — Data Science Project**

2026

R, k-Nearest Neighbors | 5,000 records | [GitHub](#)

- Built and evaluated a k-nearest neighbours classification workflow on 5,000 customer records to model loan-purchase propensity.

## **LEADERSHIP & COMMUNITY ENGAGEMENT**

---

### **HoliRun UTP — Assistant Project Director**

Jan 2026

- Served as second-in-command for a 250+ participant university fundraising colour run with a RM29,390 approved budget, coordinating an organizing committee of 30–40 students across multiple functional departments.
- Led planning meetings, monitored departmental deliverables, and coordinated preparations remotely while on exchange at Bilkent University before returning to Malaysia for event execution.

### **UTP Sangeeth — Assistant Head, Event Management**

2025 – 2026

- Coordinated meetings, manpower allocation, logistics, venue arrangements, event schedules and event flow for cultural programmes and club activities.
- Supported club recruitment, CAVE promotional activities, Pongal programmes and internal team-building initiatives.

### **EduLink: Evolving Learning Experience — Media & Promotion Committee**

Feb 2026

- Managed promotional and event-media execution for a 20-member UTP community-engagement project at SMK Bercham, Ipoh, reaching 648 students and 71 teachers.
- Produced campaign materials, social-media communications and live event coverage supporting the programme's Smart Discipline Tracker and AI Language Translator initiatives.

### **Down to Earth — Sponsorship Committee**

Mar 2024

- Contributed to sponsorship outreach for a community initiative serving children in Ipoh, supporting a Sponsorship Committee effort that secured RM4,230 in financial and in-kind contributions.

## **SCHOLARSHIPS, HONOURS & SELECTED AWARDS**

---

### **YUTP Education Grant Recipient — Yayasan Universiti Teknologi PETRONAS (YUTP)**

Sep 2023 – Present

- Merit- and need-based education grant covering full tuition fees for Foundation and undergraduate studies, with additional partial financial support for the Bilkent University exchange.

### **1st Place — Tamil Public Speaking, Kuantan District Tamil Language Carnival**

2022

Five-Star Recognition — NILAM Reading Programme, Malaysia (2022) · Third Runner-Up — Manavar Mulakkam, Astro Vaanavil, a televised Tamil-language public-speaking competition with 100+ participants (2017) · 5th Place — National Tamil Language Carnival, national participant again in 2018 (2017)

## ADDITIONAL PROFESSIONAL EXPERIENCE

---

### Horizon Advanced Tech Solutions — Sole Proprietor | Business Operations & Digital Administration

Dec 2023 – Present

Kuantan, Malaysia

- Manage financial and administrative operations for a registered project-based technical services business, including business banking, invoicing, quotation review, bookkeeping coordination and tax documentation.
- Built and maintain the company website and support digital operations, commercial documentation and selected operational and technical decisions.

### Parkson — Part-Time Sales Promoter

Apr 2023; Mar 2026

Supported retail product promotion and day-to-day sales during short-term part-time engagements.

## TECHNICAL SKILLS

---

**Programming:** Python, Java, C#, R, SQL, JavaScript, HTML/CSS

**AI & Machine Learning:** scikit-learn, Pandas, NumPy, Prophet, multimodal/VLM inference

**Document AI:** IBM Docling, Unstructured, Tesseract, PaddleOCR, Azure AI Language

**Backend & Infrastructure:** FastAPI, Celery, Redis, SQLite, REST APIs, Docker

**Development:** Git, Linux, Pytest

**Additional:** Unity, MQTT, Wireshark, Autopsy, FTK Imager

Certifications: CS50's Introduction to Artificial Intelligence with Python (HarvardX, 2026) · Microsoft Applied Skills: Build a Natural Language Processing Solution with Azure AI Language (2026)

## LANGUAGES

---

**Tamil** — Native | **English** — C1 | **Bahasa Melayu** — Fluent | **Telugu** — Basic; strong comprehension | **Mandarin Chinese** — Basic | **Turkish** — Basic

IELTS Academic — Overall 7.5 (Listening 7.5 | Reading 7.5 | Writing 6.5 | Speaking 7.5), 2026